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For the object of mass 4 kg, kinetic energy is $KE_2 = \frac{(20)^2}{2\times 4} = \frac{400}{8} = 50\,J$

Note that $KE_1 \neq KE_2$ i.e., even though both are having the same momentum, the kinetic energy of both masses is not the same. The kinetic energy of the heavier object has lesser kinetic energy than smaller mass. It is because the kinetic energy is inversely proportional to the mass ($KE \propto \frac{1}{m}$) for a given momentum.

(b) As the momentum, $p = mv$, the two objects will not have same speed.

4.2.4 Potential Energy

The potential energy of a body is associated with its position and configuration with respect to its surroundings. This is because the various forces acting on the body also depends on position and configuration.

"Potential energy of an object at a point P is defined as the amount of work done by an external force in moving the object at constant velocity from the point O (initial location) to the point P (final location). At initial point O potential energy can be taken as zero.

Mathematically, *potential energy* is defined as $U = \int \vec{F}_a . d\vec{r}$ (4.16)

where the limit of integration ranges from initial location point O to final location point P.

We have various types of potential energies. Each type is associated with a particular force. For example,

(i) The energy possessed by the body due to gravitational force gives rise to gravitational potential energy.

(ii) The energy due to spring force and other similar forces give rise to elastic potential energy.

(iii) The energy due to electrostatic force on charges gives rise to electrostatic potential energy.

We will learn more about conservative forces in the section 4.2.7. Now, we continue to discuss more about gravitational potential energy and elastic potential energy.

4.2.5 Potential energy near the surface of the Earth

The gravitational potential energy (U) at some height *h* is equal to the amount of work required to take the object from ground to that height *h* with constant velocity.

Let us consider a body of mass *m* being moved from ground to the height h against the gravitational force as shown in Figure 4.8.

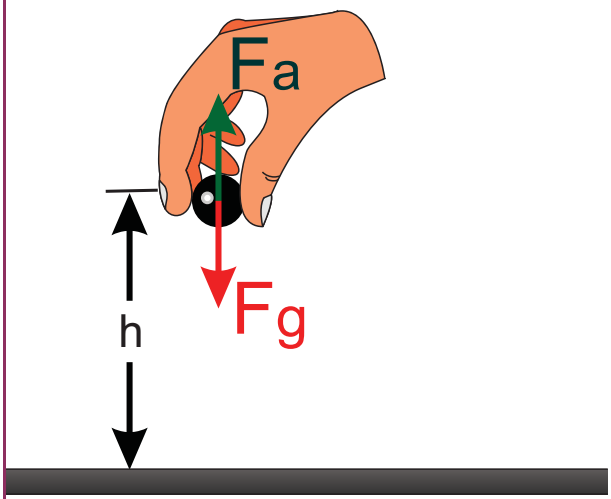


Figure 4.8 Gravitational potential energy

The gravitational force $\vec{F}_g$ acting on the body is, $\vec{F}_g = -mg\,\hat{j}$ (as the force is in y direction, unit vector $\hat{j}$ is used). Here, negative sign implies that the force is acting